

PROPERTIES OF CONTEXT-FREE LANGUAGES

a.y. 2026-2027

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Non-terminals play variables

LEMMA

Let $\mathcal{G}$ be a grammar and let $\mathcal{G}'$ be obtained from $\mathcal{G}$ by refreshing all of its non-terminals with new symbols. Then $\mathcal{L}(\mathcal{G}') = \mathcal{L}(\mathcal{G})$.

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Non-terminals play variables

EXAMPLE

- For instance, take the grammar

$$\begin{aligned} S &\rightarrow aSbC \mid ab \\ C &\rightarrow cC \mid c \end{aligned}$$

and the renaming $[R/S, D/C]$.

- Obtain the grammar

$$\begin{aligned} R &\rightarrow aRbD \mid ab \\ D &\rightarrow cD \mid c \end{aligned}$$

- Every derivation step for the first grammar can be mimicked for the second grammar, and viceversa

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Closure wrt union

LEMMA

The class of free languages is closed w.r.t. set-union.

- Meaning:
 - If $\mathcal{L}_1$ and $\mathcal{L}_2$ are free languages
 - Then $\mathcal{L}_1 \cup \mathcal{L}_2$ is a free language

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Closure wrt union

PROOF

- Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be free languages.
- Then there exist two free grammars $\mathcal{G}_1 = (V_1, T_1, S_1, \mathcal{P}_1)$ and $\mathcal{G}_2 = (V_2, T_2, S_2, \mathcal{P}_2)$ such that $\mathcal{L}_1 = \mathcal{L}(\mathcal{G}_1)$ and $\mathcal{L}_2 = \mathcal{L}(\mathcal{G}_2)$
- Also, we can assume that $\mathcal{G}_1$ and $\mathcal{G}_2$ do not share any non-terminal namely that $(V_1 \setminus T_1) \cap (V_2 \setminus T_2) = \emptyset$
- Let $\mathcal{G} = (V_1 \cup V_2 \cup \{S\}, T_1 \cup T_2, S, \mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \rightarrow S_1 \mid S_2\})$ where:
 - S is a new symbol not in $V_1 \cup V_2$
- Then $\mathcal{L}(\mathcal{G})$ is free and $\mathcal{L}(\mathcal{G}) = \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$

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Closure wrt union

PROOF: Why is $\mathcal{G}$ free?

- The productions of $\mathcal{G}$ are those in $\mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \rightarrow S_1 \mid S_2\}$
- The productions in both $\mathcal{P}_1$ and $\mathcal{P}_2$ have the form $A \rightarrow \alpha$
- The productions $S \rightarrow S_1$ and $S \rightarrow S_2$ also have the form $A \rightarrow \alpha$
- Hence $\mathcal{G}$ is free

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Closure wrt union

PROOF: Why is $\mathcal{L}(\mathcal{G}) = \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$?

- $w \in \mathcal{L}(\mathcal{G})$
- iff there exists a derivation $S \Rightarrow^* w$
- iff $S \Rightarrow S_1 \Rightarrow^* w$ or $S \Rightarrow S_2 \Rightarrow^* w$
- iff $S_1 \Rightarrow^* w$ or $S_2 \Rightarrow^* w$
- iff $w \in \mathcal{L}(\mathcal{G}_1)$ or $w \in \mathcal{L}(\mathcal{G}_2)$
- iff $w \in \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$

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Closure wrt union

PROOF: Why insisting on $\mathcal{G}_1$ and $\mathcal{G}_2$ not sharing non-terminals?

- Take

$$\begin{array}{ll} \mathcal{G}_1 : & S_1 \rightarrow aA \\ & A \rightarrow a \end{array} \qquad \mathcal{G}_2 : \begin{array}{ll} S_2 & \rightarrow bA \\ A & \rightarrow b \end{array}$$
- Then $\mathcal{L}(\mathcal{G}_1) = \{aa\}$ and $\mathcal{L}(\mathcal{G}_2) = \{bb\}$.
- However, if we just put everything together and define

$$\begin{array}{ll} \mathcal{G} : & S \rightarrow S_1 \mid S_2 \\ & S_1 \rightarrow aA \\ & S_2 \rightarrow bA \\ & A \rightarrow a \mid b \end{array}$$
- Then $\mathcal{L}(\mathcal{G}) = \{aa, ab, ba, bb\} \neq \mathcal{L}(\mathcal{G}_1) \cup \mathcal{L}(\mathcal{G}_2)$.
- What is the problem here?

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Closure wrt union

PROOF: Why insisting on $\mathcal{G}_1$ and $\mathcal{G}_2$ not sharing non-terminals?

- Take

$$\begin{array}{ll} \mathcal{G}_1 : S_1 \rightarrow aA & \mathcal{G}_2 : S_2 \rightarrow bA \\ A \rightarrow a & A \rightarrow b \end{array}$$

- Then $\mathcal{L}(\mathcal{G}_1) = \{aa\}$ and $\mathcal{L}(\mathcal{G}_2) = \{bb\}$.
- The symbol A plays distinct roles in $\mathcal{G}_1$ and in $\mathcal{G}_2$
- To reflect this distinction, rather take

$$\begin{array}{ll} \mathcal{G}_1 : S_1 \rightarrow aA & \mathcal{G}_2 : S_2 \rightarrow bA' \\ A \rightarrow a & A' \rightarrow b \end{array}$$

- No mix up of productions now, and, by construction
 $\mathcal{L}(\mathcal{G}) = \{aa, bb\}$

Closure wrt concatenation

LEMMA

The class of free languages is closed w.r.t. concatenation

- Meaning:
 - If $\mathcal{L}_1$ and $\mathcal{L}_2$ are free languages
 - Then $\{w_1w_2 \mid w_1 \in \mathcal{L}_1 \text{ and } w_2 \in \mathcal{L}_2\}$ is a free language

Closure wrt concatenation

PROOF

- Let $\mathcal{L}_1$ and $\mathcal{L}_2$ be free languages
- Then there exist two free grammars $\mathcal{G}_1 = (V_1, T_1, S_1, \mathcal{P}_1)$ and $\mathcal{G}_2 = (V_2, T_2, S_2, \mathcal{P}_2)$ such that $\mathcal{L}_1 = \mathcal{L}(\mathcal{G}_1)$ and $\mathcal{L}_2 = \mathcal{L}(\mathcal{G}_2)$
- Also, we can assume that there is no clash between the non-terminals of $\mathcal{G}_1$ and those of $\mathcal{G}_2$
- Let $\mathcal{G} = (V_1 \cup V_2 \cup \{S\}, T_1 \cup T_2, S, \mathcal{P}_1 \cup \mathcal{P}_2 \cup \{S \rightarrow S_1 S_2\})$ where S is a new symbol not in $V_1 \cup V_2$
- Then $\mathcal{L}(\mathcal{G})$ is free and $\mathcal{L}(\mathcal{G}) = \{w_1 w_2 \mid w_1 \in \mathcal{L}(\mathcal{G}_1) \text{ and } w_2 \in \mathcal{L}(\mathcal{G}_2)\}$

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Are free languages closed w.r.t. intersection?

To say something about this issue we will soon introduce the Pumping Lemma for free languages

Before that, we see how grammars can be manipulated to meet specific constraints on productions

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Chomsky normal form

LEMMA

Let $\mathcal{L}$ be a context-free language. Then there exists a context-free grammar $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \mathcal{L} \setminus \{\epsilon\}$ and $\mathcal{G}$ is such that

- ① It has no ϵ -production (i.e. no production of the shape $A \rightarrow \epsilon$)
- ② It has no unit production (i.e. no production of the shape $A \rightarrow B$)
- ③ It has no useless non-terminal (i.e. non-terminals that never appear in some derivations of some strings of terminals)
- ④ Each of its productions have
 - either the form $A \rightarrow a$
 - or the form $A \rightarrow A_1 A_2$

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Chomsky normal form

ELIMINATIONS AT WORK

$$\begin{aligned}
 S &\rightarrow ABC \mid abc \\
 A &\rightarrow aB \mid \epsilon \\
 B &\rightarrow bA \mid C \\
 C &\rightarrow \epsilon
 \end{aligned}$$

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Chomsky normal form

ϵ -PRODUCTION ELIMINATION

- Find **nullable** non-terminals, i.e., every non-terminal A such that $A \Rightarrow^* \epsilon$
- It is an inductive process
 - **Base:** if $A \rightarrow \epsilon$ is a production, then A is nullable
 - **Iteration:** if $A \rightarrow Y_1 Y_2 \dots Y_n$ is a production and $Y_1, Y_2, \dots, Y_n$ are all nullable, then A is nullable
- Substitute each production $A \rightarrow Y_1 Y_2 \dots Y_n$ by a family of productions where combinations of nullable Y_i s are removed from the body (Exception: If all of Y_i s are nullable do not take in the family the production $A \rightarrow \epsilon$)
- Eliminate every production $A \rightarrow \epsilon$

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Chomsky normal form

ϵ -PRODUCTION ELIMINATION

$$\begin{aligned} S &\rightarrow ABC \mid abc \\ A &\rightarrow aB \mid \epsilon \\ B &\rightarrow bA \mid C \\ C &\rightarrow \epsilon \end{aligned}$$

- Find **nullable** non-terminals
 - A and C nullable by $A \rightarrow \epsilon, C \rightarrow \epsilon$
 - B nullable by $B \rightarrow C$ and C nullable
 - S nullable by $S \rightarrow ABC$ and A, B, C nullable

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Chomsky normal form

ϵ -PRODUCTION ELIMINATION

$$\begin{aligned} S &\rightarrow ABC \mid abc \\ A &\rightarrow aB \mid \epsilon \\ B &\rightarrow bA \mid C \\ C &\rightarrow \epsilon \end{aligned}$$

- Substitute each production $A \rightarrow Y_1 Y_2 \dots Y_n$ by a family of productions (if all of Y_i s are nullable is useless to take in the family the production $A \rightarrow \epsilon$)

$$\begin{aligned} S &\rightarrow ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\ A &\rightarrow aB \mid a \mid \epsilon \\ B &\rightarrow bA \mid b \mid C \\ C &\rightarrow \epsilon \end{aligned}$$

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Chomsky normal form

ϵ -PRODUCTION ELIMINATION

$$\begin{aligned} S &\rightarrow ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\ A &\rightarrow aB \mid a \mid \epsilon \\ B &\rightarrow bA \mid b \mid C \\ C &\rightarrow \epsilon \end{aligned}$$

- Eliminate every production $A \rightarrow \epsilon$

$$\begin{aligned} S &\rightarrow ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \mid C \end{aligned}$$

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Chomsky normal form

USELESS NON-TERMINAL ELIMINATION

$$\begin{aligned} S &\rightarrow ABC \mid abc \mid AB \mid AC \mid BC \mid A \mid B \mid C \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \mid C \end{aligned}$$

- Useless non-terminal elimination (Here C is useless: No strings of terminals can be derived from any string containing C)

$$\begin{aligned} S &\rightarrow abc \mid AB \mid A \mid B \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \end{aligned}$$

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Chomsky normal form

UNIT PRODUCTION ELIMINATION

$$\begin{aligned} S &\rightarrow abc \mid AB \mid A \mid B \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \end{aligned}$$

- Unit production elimination

$$\begin{aligned} S &\rightarrow abc \mid AB \mid aB \mid a \mid bA \mid b \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \end{aligned}$$

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Chomsky normal form

UNIT PRODUCTION ELIMINATION

- From

$$\begin{aligned} S &\rightarrow ABC \mid abc \\ A &\rightarrow aB \mid \epsilon \\ B &\rightarrow bA \mid C \\ C &\rightarrow \epsilon \end{aligned}$$

- We get

$$\begin{aligned} S &\rightarrow abc \mid AB \mid aB \mid a \mid bA \mid b \\ A &\rightarrow aB \mid a \\ B &\rightarrow bA \mid b \end{aligned}$$

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Chomsky normal form

ONCE ALL ELIMINATIONS ARE DONE

After eliminating

- ϵ -productions
- unit productions
- useless non-terminals

We get a grammar whose productions have the form $A \rightarrow \beta$ where

- either β is a single terminal
- or β is such that $|\beta| \geq 2$

Now transform the grammar to get productions with

- either the form $A \rightarrow a$
- or the form $A \rightarrow A_1A_2$

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Chomsky normal form

LATEST TRANSFORMATION: AN EXAMPLE

Example

- $S \rightarrow aSb \mid ab$
- Both aSb and ab are to be changed. Pick up a new non-terminal for a , say A , and a new non-terminal for b , say B , and transform into
- $S \rightarrow ASB \mid AB, \quad A \rightarrow a, \quad B \rightarrow b$
- ASB is longer than 2, pick up a new non-terminal for SB , say C , and transform into
- $S \rightarrow AC \mid AB, \quad C \rightarrow SB, \quad A \rightarrow a, \quad B \rightarrow b$

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Pumping Lemma for free languages

LEMMA

Let $\mathcal{L}$ be a free language. Then

- $\exists p \in \mathbb{N}^+$ such that
- $\forall z \in \mathcal{L}$ such that $|z| > p$
- $\exists u, v, w, x, y$ such that
 - $z = uvwxy$ and
 - $|vwx| \leq p$ and
 - $|vx| > 0$ and
 - $\forall i \in \mathbb{N}. uv^iwx^iy \in \mathcal{L}$

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Pumping Lemma for free languages

PROOF

- Let $\mathcal{L}$ be a free language
- The lemma is about words longer than $p > 0$, and hence different from ϵ
- Then just consider a free grammar $\mathcal{G}$ in Chomsky normal form such that $\mathcal{L}(\mathcal{G}) = \mathcal{L} \setminus \{\epsilon\}$

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Pumping Lemma for free languages

PROOF

- Let k be the number of non-terminals in $\mathcal{G}$
- Observe that any derivation tree for words in $\mathcal{L}(\mathcal{G})$ has
 - Exactly 2^0 nodes at level 0
 - At most 2^1 nodes at level 1
 - At most 2^2 nodes at level 2
 -
 - At most 2^j nodes at level j
- Take $p = 2^{k+1}$

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Pumping Lemma for free languages

PROOF

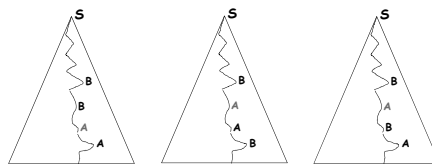
- Let $z \in \mathcal{L}$ be such that $|z| > p$
- Then the derivation tree for z must have at least $k + 2$ levels
- Then every longest path of the derivation tree for z has a terminal at the last level and traverses at least $k + 1$ non-terminals
- Hence there is at least one non-terminal occurring twice or more along the path

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Pumping Lemma for free languages

PROOF

- Consider the longest path in the tree, and the **deepest pair** of occurrences of the same non-terminal along that path (i.e. choose the non-terminal whose second occurrence is found first going bottom-up)
- For instance, below the deepest pair is always the pair of A s



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PROOF

A diagram of a triangle with a wavy line inside. The top vertex is labeled 'S'. Two points on the wavy line are labeled 'A'.

Paola Quaglia

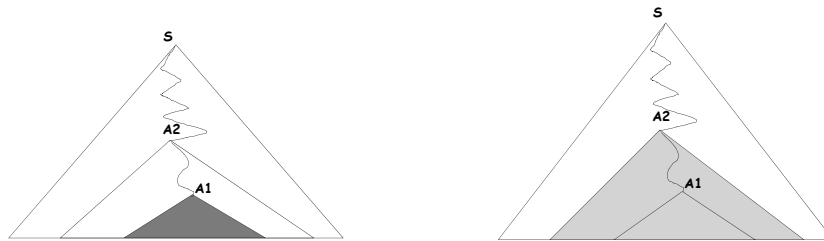
PROOF

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Pumping Lemma for free languages

PROOF

Then there are two distinct subtrees rooted at A : the “pink subtree” and the “green subtree”

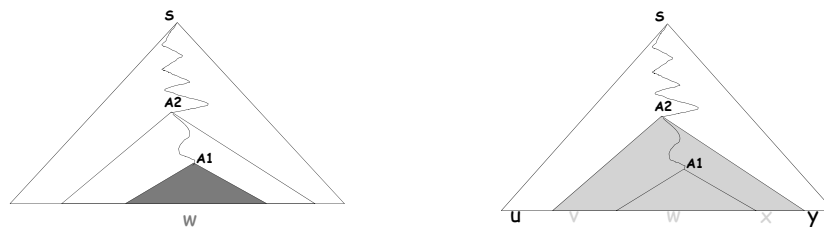


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Pumping Lemma for free languages

PROOF

Then $z = uvwx$

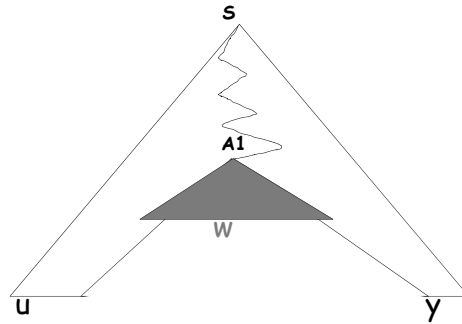


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Pumping Lemma for free languages

PROOF

Then $uv^0wx^0y \in \mathcal{L}$.

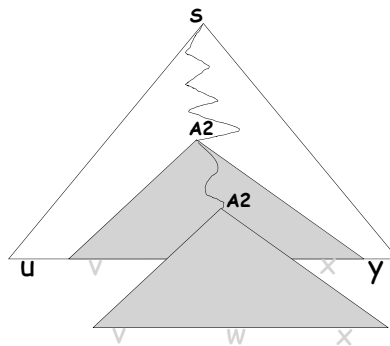


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Pumping Lemma for free languages

PROOF

Then $uv^2wx^2y \in \mathcal{L}$.

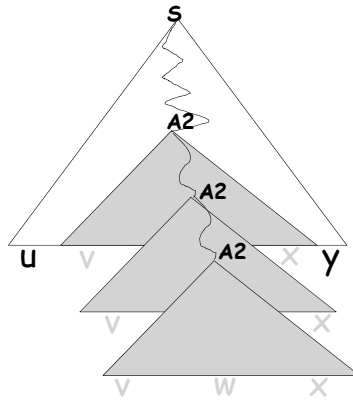


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Pumping Lemma for free languages

PROOF

Then $uv^3wx^3y \in \mathcal{L}$.



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Pumping Lemma for free languages

PROOF

Then

- $\forall i \in \mathbb{N}. uv^iwx^iy \in \mathcal{L}$
- $|vwx| \leq p$
 - Because we have chosen the deepest pair of repeated non-terminals ($A1, A2$) along the longest path from S
 - Then along the longest path from $A2$, below $A2$, no non-terminal can occur more than once
 - Then the subtree rooted at $A2$ has at most $k + 1$ levels
 - Then the length of the yield of the subtree is bound by 2^{k+1}
- $|vx| > 0$
 - Because $\mathcal{G}$ is in Chomsky normal form
 - Then it cannot be $A \Rightarrow^+ A$. It can only be $A \Rightarrow^+ \alpha A \beta$ with at least one terminal derived by either α or β

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Pumping Lemma for free languages

WHAT IS THIS LEMMA GOOD FOR?

- Recall the structure of the statement
- “ Let $\mathcal{L}$ be a free language. Then *PL-THESIS*.”
- **By no means** the lemma can be used to show that a certain language is free
- It is rather used to show that a **language is not free**
- Schema of such proofs:
 - Assume that language $\mathcal{L}$ is free
 - Show that *PL-THESIS* is false, i.e., prove *not(PL-THESIS)*
 - By contradiction, conclude that $\mathcal{L}$ is not free

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Pumping Lemma for free languages

PL-THESIS

- $\exists p \in \mathbb{N}^+. \forall z \in \mathcal{L}: |z| > p. \exists u, v, w, x, y. P$
- where
 - $P \equiv P1 \text{ and } P2 \text{ and } P3 \text{ and } P4$
 - $P1 \equiv z = uvwxy$
 - $P2 \equiv |vwx| \leq p$
 - $P3 \equiv |vx| > 0$
 - $P4 \equiv \forall i \in \mathbb{N}. uv^iwx^iy \in \mathcal{L}$

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Pumping Lemma for free languages

not(PL-THESIS)

- not ($\exists p \in \mathbb{N}^+. \forall z \in \mathcal{L}: |z| > p. \exists u, v, w, x, y. P$)
- $\forall p \in \mathbb{N}^+. \text{ not } (\forall z \in \mathcal{L}: |z| > p. \exists u, v, w, x, y. P)$
- $\forall p \in \mathbb{N}^+. \exists z \in \mathcal{L}: |z| > p. \text{ not } (\exists u, v, w, x, y. P)$
- $\forall p \in \mathbb{N}^+. \exists z \in \mathcal{L}: |z| > p. \forall u, v, w, x, y. \text{ not } (P)$
- $\forall p \in \mathbb{N}^+. \exists z \in \mathcal{L}: |z| > p. \forall u, v, w, x, y. \text{ not } (P1 \text{ and } P2 \text{ and } P3 \text{ and } P4)$

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Pumping Lemma for free languages

not ($P1$ and $P2$ and $P3$ and $P4$)

- not ($P1$ and $P2$ and $P3$ and $P4$)
- not (($P1$ and $P2$ and $P3$) and $P4$)
- not ($P1$ and $P2$ and $P3$) or not $P4$
- ($P1$ and $P2$ and $P3$) implies not $P4$
- ($P1$ and $P2$ and $P3$) implies not ($\forall i \in \mathbb{N}. uv^i wx^i y \in \mathcal{L}$)
- ($P1$ and $P2$ and $P3$) implies $\exists i \in \mathbb{N}. \text{ not } (uv^i wx^i y \in \mathcal{L})$

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Pumping Lemma for free languages

not(PL-THESIS)

- $\forall p \in \mathbb{N}^+. \exists z \in \mathcal{L}: |z| > p. \forall u, v, w, x, y.$
 - $(z = uvwxy \text{ and } |vwx| \leq p \text{ and } |vx| > 0)$
 - implies
 - $\exists i \in \mathbb{N}. uv^iwx^iy \notin \mathcal{L}$
- Operationally:
- **Whichever** positive natural number p is
- Carefully **choose** a word z longer than p and belonging to the language such that
- **Whichever** unpacking of z into $uvwxy$ with $|vwx| \leq p$ and $|vx| > 0$ is taken
- A natural number i can be **chosen** which is such that $uv^iwx^iy \notin \mathcal{L}$

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Pumping Lemma at work

$$\begin{aligned} \mathcal{G}: \quad S &\rightarrow aSBc \mid abc \\ cB &\rightarrow Bc \\ bB &\rightarrow bb \end{aligned}$$

- $\mathcal{G}$ is context-dependent and $\mathcal{L}(\mathcal{G}) = \{a^n b^n c^n \mid n > 0\}$.
- Is $\mathcal{L}(\mathcal{G})$ a free language?

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Pumping Lemma at work

LEMMA

$\mathcal{L} = \{a^n b^n c^n \mid n > 0\}$ is not free

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Pumping Lemma at work

PROOF

- Suppose $\mathcal{L}$ is free, and let p be an arbitrary positive integer
- Take $z = a^p b^p c^p$
- If ($z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$)
- Then vx cannot have occurrences of both a s and c s because the last occurrence of a and the first occurrence of c are $p + 1$ positions far. In fact, for some positive k and j
 - Either $vwx = a^k$
 - Or $vwx = a^k b^j$
 - Or $vwx = b^j$
 - Or $vwx = b^j c^k$
 - Or $vwx = c^k$

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Pumping Lemma at work

PROOF

- Then vx has either no occurrences of c or no occurrences of a
- Then uv^0wx^0y cannot have the form $a^n b^n c^n$, hence $uv^0wx^0y \notin \mathcal{L}$
- Then, by contradiction wrt the Pumping Lemma, $\mathcal{L}$ is not free

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Pumping Lemma at work

AGAIN ON THE PROOF STRUCTURE

- ... let p be an arbitrary positive integer
 $\forall p$: Any p
- Take $z = a^p b^p c^p$
 $\exists z$: Choose $z \in \mathcal{L}$ longer than p
- ... If ($z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$) then
 $\forall u, v, w, x, y : z = uvwxy$ and $|vwx| \leq p$ and $|vx| > 0$
- ... $uv^0wx^0y \notin \mathcal{L}$
 $\exists i$: Choose a value for the iterator
- ... contradiction

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Pumping Lemma at work

$$\begin{aligned}
 \mathcal{G}: \quad S &\rightarrow CD \\
 C &\rightarrow aCA \mid bCB \mid \epsilon \\
 AD &\rightarrow aD \\
 BD &\rightarrow bD \\
 Aa &\rightarrow aA \\
 Ab &\rightarrow bA \\
 Ba &\rightarrow aB \\
 Bb &\rightarrow bB \\
 D &\rightarrow \epsilon
 \end{aligned}$$

- $\mathcal{G}$ is context-dependent
- What is $\mathcal{L}(\mathcal{G})$?

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Pumping Lemma at work

- Derived strings have a bookmark D initially at rightmost position

$$S \rightarrow CD$$

- Strings can only grow longer by replacing C

$$C \rightarrow aCA \mid bCB \mid \epsilon$$

- Non-terminals close to the rightmost delimiter can be converted to the corresponding terminal

$$AD \rightarrow aD$$

$$BD \rightarrow bD$$

- Non-terminals and terminals can be swapped when the terminal is at the right of the non-terminal

$$Aa \rightarrow aA$$

$$Ab \rightarrow bA$$

$$Ba \rightarrow aB$$

$$Bb \rightarrow bB$$

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Pumping Lemma at work

S	by $S \rightarrow CD$
$\Rightarrow CD$	by $C \rightarrow aCA$
$\Rightarrow aCAD$	by $C \rightarrow aCA$
$\Rightarrow aaCAAD$	by $C \rightarrow bCB$
$\Rightarrow aabCBAAD$	by $C \rightarrow \epsilon$
$\Rightarrow aabBAAD$	by $AD \rightarrow aD$
$\Rightarrow aabBAaD$	by $Aa \rightarrow aA$
$\Rightarrow aabBaAD$	by $Ba \rightarrow aB$
$\Rightarrow aabaBAD$	by $AD \rightarrow aD$
$\Rightarrow aabaBaD$	by $Ba \rightarrow aB$
$\Rightarrow aabaaBD$	by $BD \rightarrow bD$
$\Rightarrow aabaabD$	by $D \rightarrow \epsilon$
$\Rightarrow aabaab$	

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Pumping Lemma at work

$$\mathcal{G} : \begin{array}{lcl} S & \rightarrow & CD \\ C & \rightarrow & aCA \mid bCB \mid \epsilon \\ AD & \rightarrow & aD \\ BD & \rightarrow & bD \\ Aa & \rightarrow & aA \\ Ab & \rightarrow & bA \\ Ba & \rightarrow & aB \\ Bb & \rightarrow & bB \\ D & \rightarrow & \epsilon \end{array}$$

- $\mathcal{L}(\mathcal{G}) = \{ww \mid w \in \{a,b\}^*\}$
- Is $\mathcal{L}(\mathcal{G})$ a free language?

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Pumping Lemma at work

LEMMA

$\mathcal{L}(\mathcal{G}) = \{ww \mid w \in \{a, b\}^*\}$ is not free

Proof

Analogous to the previous one.

A good choice for z is $z = a^p b^p a^p b^p$

Free or not?

- $\{a^n b^n c^n \mid n > 0\}$
- Not free, by previous lemma

- $\{a^n b^n c^j \mid n, j > 0\}$
- Free, concatenation of two free languages
- $\{a^n b^n \mid n > 0\}$ and $\{c^j \mid j > 0\}$

- $\{a^j b^n c^n \mid j, n > 0\}$
- Free, concatenation of two free languages
- $\{a^j \mid j > 0\}$ and $\{b^n c^n \mid n > 0\}$

Closure wrt intersection does not hold

LEMMA

The class of free languages is not closed w.r.t. intersection.

Proof

- By contradiction
- Take two free languages $\mathcal{L}_1$ and $\mathcal{L}_2$ whose intersection is not free
- $\mathcal{L}_1 = \{a^n b^n c^j \mid n, j > 0\}$
- $\mathcal{L}_2 = \{a^j b^n c^n \mid n, j > 0\}$
- $\mathcal{L}_1 \cap \mathcal{L}_2 = \{a^n b^n c^n \mid n > 0\}$

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Training

$$\mathcal{G}: S \rightarrow aSc \mid aTc \mid T$$

$$T \rightarrow bTa \mid ba$$

- Is $\mathcal{G}$ ambiguous?
- Yes
 - $S \Rightarrow aTc \Rightarrow abac$
 - $S \Rightarrow aSc \Rightarrow aTc \Rightarrow abac$
- What is $\mathcal{L}(\mathcal{G})$?

$$\{a^k b^n a^n c^k \mid k \geq 0, n > 0\}$$

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Training

$$\mathcal{G}: S \rightarrow 0B \mid 1A$$

$$A \rightarrow 0 \mid 0S \mid 1AA$$

$$B \rightarrow 1 \mid 1S \mid 0BB$$

- What is $\mathcal{L}(\mathcal{G})$?

$$\{w \mid w \in \{0,1\}^* \text{ and } \#(0,w) = \#(1,w)\}$$

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Training

- Define $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \{a^k b^n c^{2k} \mid k, n > 0\}$

$$S \rightarrow aScc \mid aBcc$$

$$B \rightarrow bB \mid b$$

- Define $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G}) = \{a^k b^n c^{2k} \mid k, n \geq 0\}$

$$S \rightarrow aScc \mid B$$

$$B \rightarrow bB \mid \epsilon$$

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Training

$$\mathcal{G} : \quad S \rightarrow aBS \mid bA$$

$$aB \rightarrow Ac \mid a$$

$$bA \rightarrow S \mid Ba$$

- Is $\mathcal{L}(\mathcal{G}) = \emptyset$?

No: $\underline{S} \Rightarrow aB\underline{S} \Rightarrow \underline{aB}bA \Rightarrow \underline{abA} \Rightarrow \underline{aBa} \Rightarrow aa$

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Training

- Define a grammar $\mathcal{G}$ such that $\mathcal{L}(\mathcal{G})$ is the set of all the even binary numbers

$$S \rightarrow 0S \mid 1S \mid 0$$

- Define a grammar $\mathcal{G}'$ such that $\mathcal{L}(\mathcal{G}') = \{1^n 0 \mid n \geq 0\}$

$$S \rightarrow 1S \mid 0$$

- Is $\mathcal{L}(\mathcal{G}') = \mathcal{L}(\mathcal{G})$?

No: 000000 is in $\mathcal{L}(\mathcal{G})$ but not in $\mathcal{L}(\mathcal{G}')$

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